Unit 2 – Exploring Derivatives

THE CONSTANT RULE

If $f(x)=c$, c is a constant, then $f'(x)=0$.

$f(x)=c$

$f(3)=c$

$f(x)=c$

$f(x+1)=c$

$f(\text{mrv})=c$

$f(67)=c$

Proof:

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

Lagrange

$$=\lim_{h\to 0}\frac{c-c}{h}$$

$$=\lim_{h\to 0}\frac{0}{h}=\lim_{h\to 0} 0$$

$$=0$$

Example: Determine the derivative for each of the following.

a. $f(x)=0$ b. $f(x)=\sqrt{7}$ c. $y=-\frac{1}{2}$ d. $y=-4.67$

$f'(x)=0$ $f'(x)=0$ $y'=0$ $\frac{dy}{dx}=0$

$f(x)=\pi^3$

$f'(x)\neq 3\pi^2$ | Not power rule

$f(x)=x^4$
$f'(x)=4x^3$

THE POWER RULE

If $f(x)=x^n$, where $n\in\mathbb{R}$, then $f'(x)=nx^{n-1}$.

Proof:

Let $f(x)=x^n$

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$=\lim_{h\to 0}\frac{(x+h)^n-x^n}{h}$$

$$=\lim_{h\to 0}\frac{\cancel{x^n}+nx^{n-1}(h)+\frac{n(n-1)}{2}x^{n-2}(h^2)\ldots+(h)^n\cancel{-x^n}}{h}$$

Factor out the "h"

$$=\lim_{h\to 0}\frac{\cancel{h}[nx^{n-1}+\frac{n(n-1)}{2}x^{n-2}(h)\ldots+(h^{n-1})]}{\cancel{h}}$$

Let "h" go to zero

$$=nx^{n-1}$$

Example: Determine the derivative for each of the following.

a. $f(x)=x$

$f'(x)=1\cdot x^{1-1}$

$f'(x)=1x^0$

$=1$

b. $f(x)=x^5$

$f'(x)=5x^4$

c. $y=\sqrt{x}$

$y=x^{\frac{1}{2}}$

$y'=\frac{1}{2}x^{-\frac{1}{2}}$

$=\frac{1}{2\sqrt{x}}$

d. $y=x^{-4}$

$\frac{dy}{dx}=-4x^{-5}$

$=\frac{-4}{x^5}$

e. $f(x)=\frac{1}{\sqrt[3]{x}}$

$f(x)=x^{-\frac{1}{3}}$

$f'(x)=-\frac{1}{3}\cdot x^{-\frac{4}{3}}$

$=\frac{-1}{3\sqrt[3]{x^4}}=\frac{-1}{3x^{4/3}}$

Both ok!

f. $f(x)=\sqrt[6]{x}$

$x^{\frac{1}{6}}$

$f'(x)=\frac{1}{6}x^{-\frac{5}{6}}$

$=\frac{1}{6\cdot\sqrt[6]{x^5}}$

$=\frac{1}{6x^{5/6}}$

$\lim_{h\to 0} c\cdot g(x)$

$= c \lim_{h\to 0} g(x)$

THE CONSTANT MULTIPLE RULE

If $f(x)=c\cdot g(x)$, c is a constant, then $f'(x)=c\cdot g'(x)$.

Example: Determine the derivative for each of the following.

a. $f(x)=4x^3$

$f'(x) = 12x^2$

$f'(x) = 4\cdot(3x^2)$

$= 12x^2$

b. $f(x)=-16x^{-4}$

$f'(x) = 64x^{-5}$

$= \frac{64}{x^5}$

c. $y=-2x^{\frac{3}{2}}$

$\frac{dy}{dx} = -3x^{\frac{1}{2}}$

$= -3\sqrt{x}$

d. $y=\sqrt{6}x^{-\frac{3}{5}}$

$y' = \frac{-3\sqrt{6}}{5} x^{-\frac{8}{5}}$

$= \frac{-3\sqrt{6}}{5\cdot\sqrt[5]{x^8}}$

$= \frac{-3\sqrt{6}}{5\cdot x^{\frac{8}{5}}}$

e. $y=3x^{\sqrt{5}}$

$\frac{dy}{dx} = 3\sqrt{5}\,x^{\sqrt{5}-1}$

Leibniz.

leave exact

f. $y=\pi x^{\pi}$

$y' = \pi^2\cdot x^{\pi-1}$

$y' = \pi\cdot\pi\, x^{\pi-1}$

$= \pi^2 x^{\pi-1}$

$y = \pi^3$

$y' = 0$

$3\pi^2$ ~~X~~

SUM & DIFFERENCE RULES

If both $p(x)$ and $q(x)$ are differentiable and $f(x) = p(x) \pm q(x)$, then $f'(x) = p'(x) \pm q'(x)$.

Example: Determine the derivative for each of the following.

a. $f(x) = \sqrt{\frac{2}{x}} + \sqrt{\frac{x}{3}}$

$$f(x) = \sqrt{2}\, x^{-\frac{1}{2}} + \frac{x^{\frac{1}{2}}}{\sqrt{3}}$$

$$f'(x) = -\frac{\sqrt{2}}{2} x^{-\frac{3}{2}} + \frac{x^{-\frac{1}{2}}}{2\sqrt{3}}$$

$$= \frac{-\sqrt{2}}{2\sqrt{x^3}} + \frac{1}{2\sqrt{3x}}$$

b. $y = -\frac{\sqrt{2}}{2} x^{-\frac{3}{2}} + \frac{x^{-\frac{1}{2}}}{2\sqrt{3}}$

$$y' = \frac{3\sqrt{2}}{4} x^{-\frac{5}{2}} - \frac{x^{-\frac{3}{2}}}{4\sqrt{3}}$$

$$= \frac{3\sqrt{2}}{4\sqrt{x^5}} - \frac{1}{4\sqrt{3x^3}}$$

c. $y = x^{\pi} - \pi x^3$

$$y' = \pi x^{\pi - 1} - 3\pi x^2$$

$$y' = \pi\left(x^{\pi-1} - 3x^2\right)$$

d. $f(x) = 7x^4 - \sqrt{x} + 3x^{\frac{3}{2}} - 401$

$x^{\frac{1}{2}}$

$$f'(x) = 28x^3 - \frac{1}{2\sqrt{x}} + \frac{9}{2}\sqrt{x}$$

Example: Determine the equation of the tangent to the graph of $y = x^3 + 2x^2 - 4x + 1$ at $x = 4$.

$\frac{dy}{dx} = 3x^2 + 4x - 4$

$y - y_1 = m(x - x_1)$

$y - 81 = 60(x - 4)$

$m_t\Big|_{x=4} = 48 + 16 - 4$

$= 60$

Example: Determine the values(s) of x so that the tangent to $f(x) = \frac{3}{\sqrt[3]{x}}$ is parallel to the line $x + 16y + 3 = 0$.

$f(x) = 3 \cdot x^{-\frac{1}{3}}$

$f'(x) = -x^{-\frac{4}{3}}$

$= -\frac{1}{\sqrt[3]{x^4}}$

$x + 16y + 3 = 0$

$y = \frac{-1}{16}x - \frac{3}{16}$

$\therefore m_t = \frac{-1}{16}$

$\therefore \frac{-1}{\sqrt[3]{x^4}} = \frac{-1}{16}$

$\Rightarrow \sqrt[3]{x^4} = 16 \rightarrow x^{\frac{4}{3}} = 16 \rightarrow x = 16^{\frac{3}{4}}$

$\therefore x = \pm 8$

Additional Questions

1. Determine the derivative of each of the following.

a) $f(x) = -\frac{3}{4}\sqrt[4]{x^5} - \frac{4}{3\sqrt{x^3}} + \pi^2 x^3 - \frac{7}{3}$

$f(x) = -\frac{3}{4}x^{\frac{5}{4}} - \frac{4}{3}x^{-\frac{3}{2}} + \pi^2 x^3 - \frac{7}{3}$

$f'(x) = -\frac{15}{16}x^{\frac{1}{4}} + 2x^{-\frac{5}{2}} + 3\pi^2 x^2$

$= \frac{-15\sqrt[4]{x}}{16} + \frac{2}{\sqrt{x^5}} + 3\pi^2 x^2$

b) $g(x) = 4\sqrt[4]{x^3}\left(\pi\sqrt[5]{x} - 2^3\pi\right)$

$g(x) = 4\pi x^{\frac{19}{20}} - 32\pi x^{\frac{3}{4}}$

$g'(x) = \frac{19}{5}\pi x^{-\frac{1}{20}} - 24\pi x^{-\frac{1}{4}}$

$= \frac{19\pi}{5\sqrt[20]{x}} - \frac{24\pi}{\sqrt[4]{x}}$

c) $g(x) = \frac{3x^4 + 2\pi x^3 - 5\sqrt[3]{x}}{4x^2}$

$g(x) = \frac{3x^2}{4} + \frac{\pi x}{2} - \frac{5}{4}x^{-\frac{5}{3}}$

$g'(x) = \frac{3x}{2} + \frac{\pi}{2} + \frac{25}{12}x^{-\frac{8}{3}}$

$= \frac{3x}{2} + \frac{25}{12\sqrt[3]{x^8}} + \frac{\pi}{2}$